Backward

Backward retiming pushes registers toward inputs , increase r(v) on the chosen node

The idea

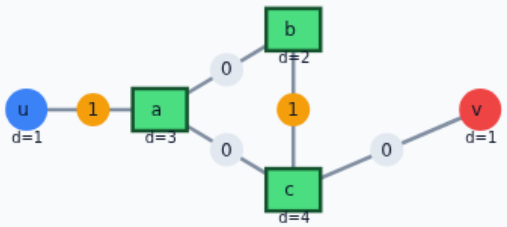
- Backward_retime('c', r, 1) sets r(c)=+1; e3 w_r = 0+0−1 = −1 violates w_r ≥ 0

Backward +r

BACKWARD RETIME

Backward +r

Step 1 / 5 · def · module02-05-backward-retime



Explanation

Backward retiming increases r(v): registers move toward inputs across v.

- +r(v)

```
toward PI
```

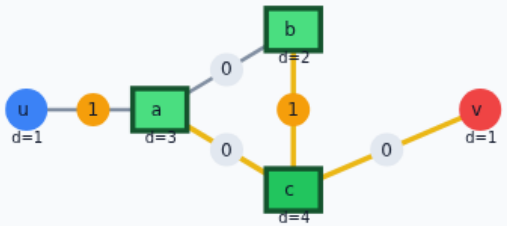
Backward +r

Pick c

BACKWARD RETIME

Pick c

Step 2 / 5 · pick · module02-05-backward-retime



Explanation

Try backward on c (the slow comb, d=4).

- node c

d=4

Pick c

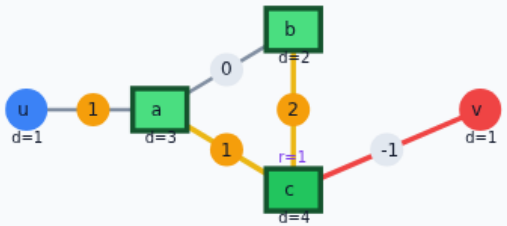
Apply +1

BACKWARD RETIME

Apply +1

Step 3 / 5 · apply · module02-05-backward-retime

legal=false P=5



Explanation

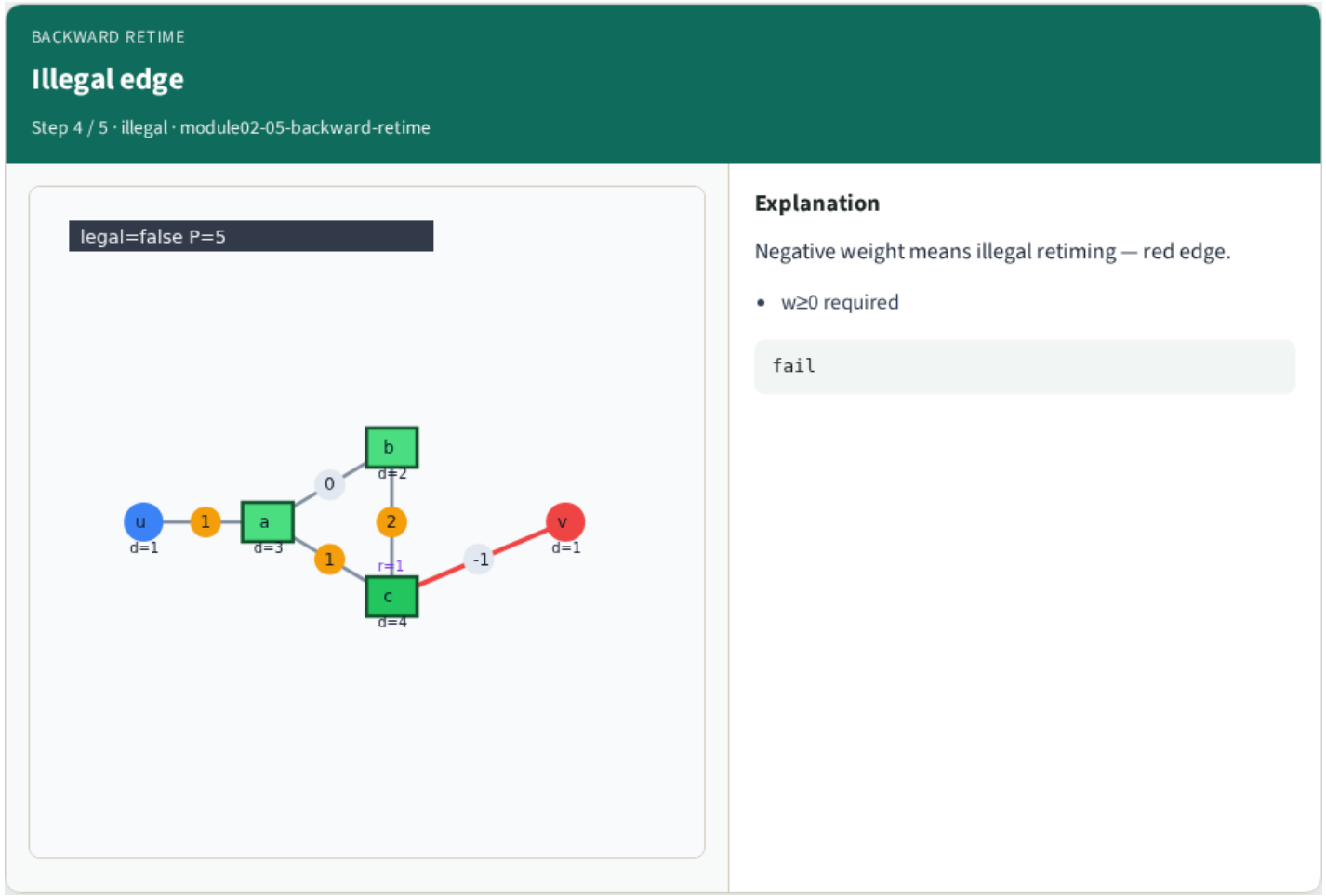
r(c)=+1 changes outbound e3 weight to −1.

- w_r(e3)

```
danger
```

Apply +1

Illegal edge



Illegal edge

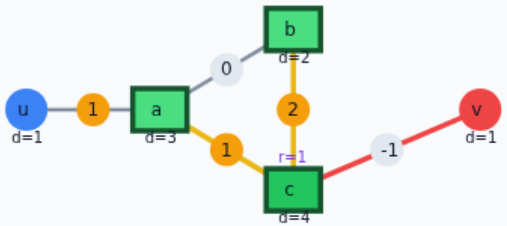
Not every move

BACKWARD RETIME

Not every move

Step 5 / 5 · lesson · module02-05-backward-retime

legal=false P=5



Explanation

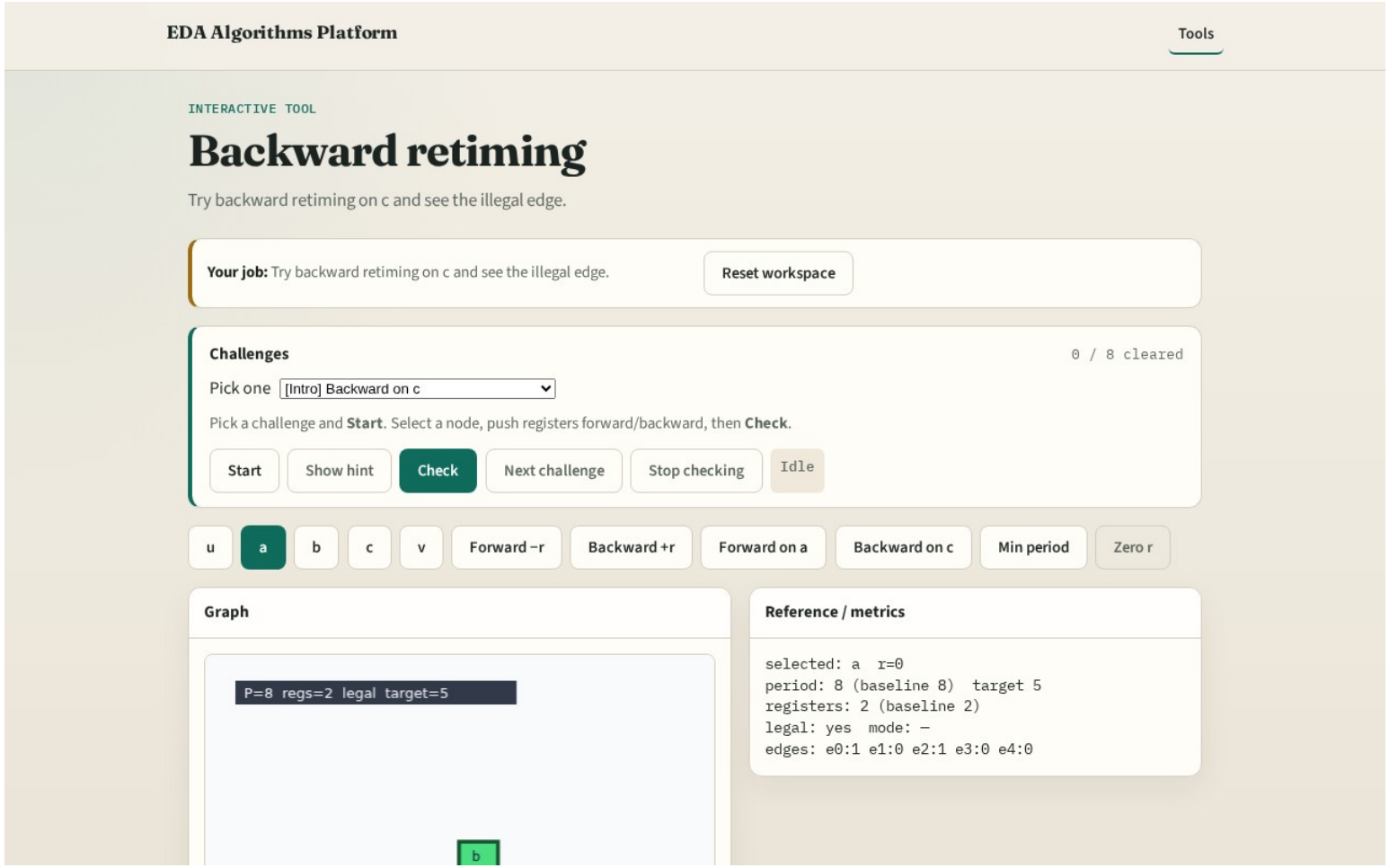
Backward is valid only when every w_r stays non-negative.

- legality

```
Next lab
```

Not every move

Browser lab track



Browser lab starter

Implement track

- Call backward_retime('c', r, 1); assert not is_legal(edges, r)

Pitfalls

- Assuming backward is always safe
- Ignoring PO-adjacent edges

Your turn

- Next: legal retiming check for all edges